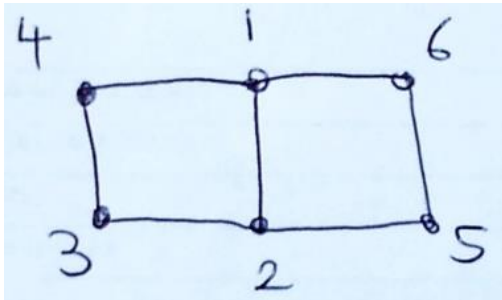


Discrete Mathematics: Graph Theory revision

(a)



(b) 4-1-6-5-2-3-4

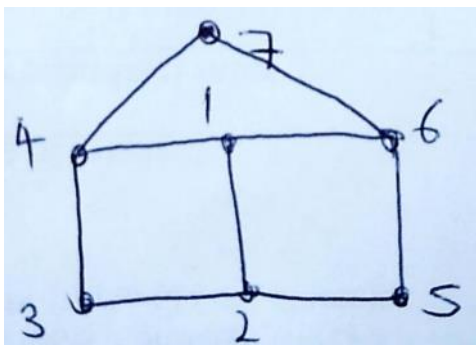
(c) 1-6-5-2-1-4-3-2

(d) There are two vertices of odd degree (1 and 2). For an Eulerian circuit to exist, all vertices must have even degree.

(e) See below for the graph.

1-2-3-4-7-6-5 is a Hamiltonian path.

Suppose there were a Hamiltonian cycle, because vertices 3, 5, and 7 have degree 2 both edges incident at those vertices must be in the cycle. But this then describes a cycle (7-6-5-2-3-4-7) that excludes 1. Contradiction.

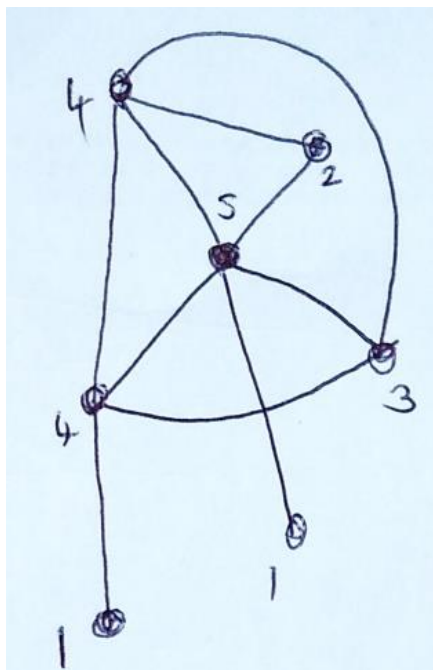


(f) For a Eulerian trail there can be at most two vertices of odd degree. This graph has 4 (1, 2, 4, and 6).

2 (a) A vertex of degree 2 can only be adjacent to two other vertices, therefore some of the vertices of degree 2 are not adjacent. The sum of the degrees of these non-adjacent vertices is 4, which is less than the number of vertices in the graph (5). Therefore Ore's condition is not satisfied.

(b) The only way to have this degree sequence is to have a 5-cycle, which IS a Hamiltonian cycle.

3 (a)

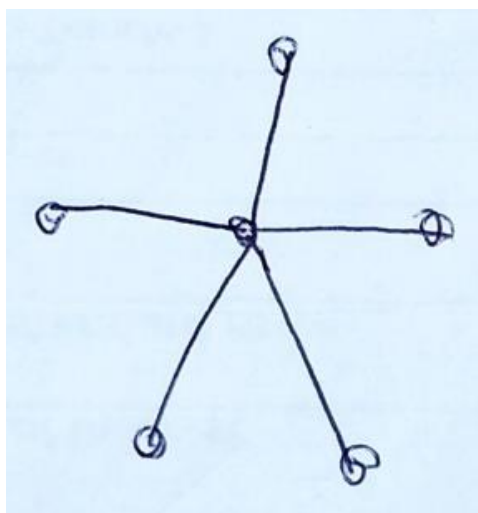


(b) Three vertices of odd degree. Graphs must have an even number of vertices of odd degree because the sum of the degrees is twice the number of edges.

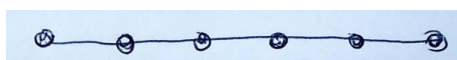
4 By definition, a tree contains no cycles.

5 A connected tree has six vertices. Draw examples in which the largest vertex degree is:

(a)



(b)



6 (a) $\frac{6 \times 5}{2} = 15$ edges

(b) We need every vertex to have even degree, so the maximum degree is 4. If all six vertices have degree 4 then the number of edges is $6 \times 4 \div 2 = 12$. This is enough to show that we cannot have more than 12 edges, but it doesn't guarantee that 12 is an attainable maximum. Here is how to create such a graph: Start with the complete graph, in which every edge has degree 5. If we pair up the vertices and remove the edge joining each pair then every vertex has degree 4 and the graph will have 12 edges.

7 There are 16. One option would be to list them all, but a combinatorial argument is quicker: There are 6 edges, and a spanning tree will use exactly 3 edges, of which there are $6C3 = 20$ different selections. Which of these selections do not describe a spanning tree? The only way that three edges can fail to be a tree is if they form a cycle. There are four possible cycles in the graph (corresponding to the $4C3 = 4$ selections of three vertices). So the remaining 16 selections must be trees, and hence spanning trees.

8 There are 6 vertices of odd degree, so there is no Eulerian trail in this graph. Imagine modifying the graph by adding extra edges. If an additional edge is added between two of the three odd vertices on the left, and an additional edge is added between two of the three odd vertices on the right, then our new (non-simple) graph will only have two odd vertices (one on the left, and one on the right). We can now go over each edge exactly once, starting at one odd vertex and finishing at the other. In doing this we have effectively 'retraced' two of the edges.

9 Looking at the degree sequences:

- (a) 4,4,3,3,3,2
- (b) 4,4,4,4,3,3,2
- (c) 4,4,4,4,3,3,2
- (d) 4,4,4,4,3,3,2
- (e) 4,4,4,4,4,3,3
- (f) 4,4,4,4,4,3,3

Therefore graph (a) cannot be isomorphic to any of the other graphs. It is possible that some of (b), (c), or (d) could be isomorphic to each other, and it is possible that (e) and (f) are isomorphic.

Graphs (b) and (c) are not isomorphic. There are lots of differences you could remark on, but I noticed that in (b) the vertex of degree 2 (C) is adjacent to a vertex of degree 3 (F) whereas in (c) the vertex of degree 2 (D) is adjacent only to vertices of degree 4 (C and E).

Graph (b)	Graph (d)
A	D
B	E
C	A
D	C
E	G
F	B
G	F

Graphs (b) and (d) are isomorphic. To find the correspondence start with the vertex of degree 2 and be systematic:

(e) and (f) are not isomorphic. In (f) there is a path of length three from one degree vertex to the other (F-B-E). In (e) the shortest path between the degree 3 vertices (D and F) has length 4.

- 10 (a) All contain at least two vertices of odd degree, and so none are Eulerian.
- (b) All contain at least two vertices of odd degree, and so none are Eulerian.
- (c) All of the graphs contain a Hamiltonian cycle.
- (d) In the first four graphs the vertex of degree 2 is non-adjacent to at least one vertex of degree 4 (or 3). Because $2+4=6<7$ (or $2+3=5<7$) and there are 7 vertices, these graphs do not satisfy Ore's condition.

In graphs (e) and (f) there are only vertices of degree 4 and 3. The two vertices of degree 3 are adjacent, so there do not exist non-adjacent vertices in which the sum of the degrees is less than the number of vertices. **So (e) and (f) satisfy the condition of Ore's Theorem.**

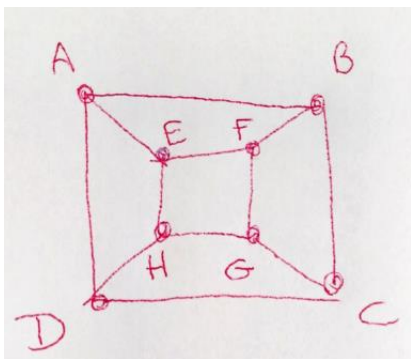
- 11 A bipartite graph cannot contain any 3-cycles, and A-C-F-A and D-E-G-D are completely separate 3-cycles. Therefore we will need to remove at least one edge from each of these cycles, so a minimum of two edges will need to be removed.

Noticing that A-C-D-A is also a 3-cycle suggests that we should remove AD.

With perhaps a bit of experimenting (try to 2-colour the graph starting at A with AD deleted) we realise that also removing EG is sufficient.

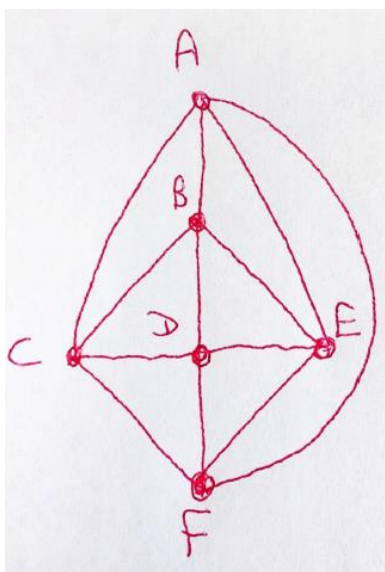
So remove AD and EG and the graph is bipartite. One of the bipartite sets is {A, C, E, G} and the other is {B, D, F}.

- 12 (a) The idea is to draw a graph showing the vertices of a cube. Two vertices are adjacent if and only if there is an edge between them. An example is:



- (i) Then A-B-C-D-H-G-F-E-A is an example of a Hamiltonian cycle.
- (ii) All the vertices have degree 3 (the graph is 3-regular). There are more than two vertices of odd degree, so no Eulerian trail.

- (b) The idea here is to draw a graph where the vertices are the faces of a cube. Two vertices are adjacent if the faces they represent share an edge. For example:



- (i) A-B-C-D-E-F-A is an example
- (ii) All the of the graph have degree 4 (the graph is 4-regular), so a Eulerian circuit exists in the graph. This corresponds to Silas crossing every edge exactly once, and finishing on the same face on which he started.
- 13 Suppose that it is possible. Draw a graph in which the vertices represent the 9 lines. Two vertices are adjacent if and only if the lines cross. All the vertices will have degree 3 (the graph is 3-regular). But there are an odd number of vertices, so this is not possible. Contradiction.